

# LUYANG HAN

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## EDUCATION

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### Columbia University

Sep 2025 – May 2027 (expected)

MA in Statistics (Machine Learning track), Research Specialization · GPA: 3.91/4.00

### Xi'an Jiaotong-Liverpool University (XJTLU)

Sep 2021 – Jul 2025

Bachelor of Science (Honours) in Financial Mathematics (awarded by University of Liverpool)

Bachelor of Economics in Financial Mathematics (awarded by XJTLU) · WES GPA: 3.68/4.00

## PUBLICATIONS

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**Han, L., & Di, X.** Cross-Classified Random-Effects Difference-in-Differences Model: A Case Study for New York City's Congestion Pricing. *Transportation Research Board (TRB) Annual Meeting 2027* (accepted for presentation; under review at *Transportation Research Record*) · [\[PDF\]](#)

## RESEARCH INTERESTS

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Probabilistic Machine Learning for Origin–Destination (OD) Data · Causal Inference for Transportation Policy · Spatio-Temporal Mobility Modeling

## RESEARCH EXPERIENCE

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### Uncovering Significant Cross-Scale Flow Clusters in NYC Multimodal OD Data

Columbia University

Jul 2026 – Present

Graduate Researcher · Advisor: Prof. Xuan Sharon Di

- Targeting recurrent, cross-scale OD flow clusters that stand out from day-to-day noise in subway, ride-hail, taxi and bike-share trips across NYC
- Reimplemented the XFlowMap scan-statistic method from scratch as a frequentist baseline; developing a Bayesian hierarchical counterpart that pools information across neighboring OD pairs and selects significant clusters by posterior exceedance probabilities under Bayesian false discovery rate (FDR) control
- Developing an attribution step that explains detected clusters with multiple OD features, distinguishing different types of flows such as AM-peak and PM-peak commuting

### NYC Congestion Pricing Evaluation

Columbia University

Nov 2025 – Aug 2026

Graduate Researcher · Advisor: Prof. Xuan Sharon Di

- Estimated the first-year causal effect of NYC's congestion pricing on ridership of five modes (Yellow Taxi, Uber, Lyft, Subway, Citi Bike), decomposed by travel direction, time window and distance from the tolled zone
- Harmonized two years (2024–2025) of trip records into comparable OD panels, grouping neighborhoods into near and far rings by Ward hierarchical clustering; proposed a cross-classified random-effects difference-in-differences model, estimated by REML and validated against nested two-way fixed-effects models
- Untolled-mode ridership rose only on commute-direction flows near or inside the zone (Subway inbound in the morning, Citi Bike outbound in the evening), while tolled-mode ridership fell across all rings and time windows; existing commute patterns governed the mechanism by which the charge took effect

### Road-Segment Bus Speed Estimation from Real-Time and Static GTFS Data

Columbia University

Apr 2026

Independent Project

- Estimated road-segment, minute-level bus operating speeds across NYC from real-time automatic vehicle location (AVL) feeds, as a performance measure for transit service and policy evaluation
- Built an end-to-end pipeline that matched 100% of real-time trips to static GTFS schedules, map-matched GPS points to route shapes by projection, filtered implausible inter-point speeds, and merged overlapping route segments into a deduplicated road network
- The network-wide mean speed was within 7% of MTA's published bus speeds and the estimated speed distribution closely matched MTA's official distribution; results can be aggregated to flexible spatial and temporal resolutions · [\[Visualization\]](#)

## Goodness-of-Fit Tests for Conditional Moment Models in Reproducing Kernel Hilbert Space (RKHS)

XJTLU, Final Year Project

Oct 2024 – May 2025

Undergraduate Researcher · Advisor: Dr. Yuhao Li

- Developed specification tests of whether the conditional mean of a response follows a hypothesized parametric form, robust to the choice of parameter estimator
- Constructed kernel-based integrated conditional moment (ICM) test statistics in an RKHS, derived their asymptotic null distributions via U-statistics and Mercer expansions, and proposed a modified Gaussian kernel that removes the first-order parameter-estimation effect
- In Monte Carlo simulations with  $n = 1000$ , empirical sizes of 0.098, 0.034 and 0.006 were close to the nominal 10%, 5% and 1% levels, and power reached 1.00 against all five nonlinear alternatives · [\[Thesis PDF\]](#)

## Mean-CVaR-Skewness Portfolio Optimization under the Generalized Hyperbolic Distribution

XJTLU, Summer Undergraduate Research Fellowship (SURF)

May 2024 – Jul 2024

Undergraduate Researcher · Advisor: Dr. Arodh Lal Karn

- Developed a portfolio optimization approach that controls tail risk while accounting for skewness under heavy-tailed, asymmetric returns
- Fit a multivariate Generalized Hyperbolic (GH) distribution to returns of 30 U.S. stocks via the EM algorithm; derived a closed-form approximation of portfolio CVaR, and closed-form portfolio skewness from third moments of the generalized inverse Gaussian (GIG) mixing variable
- Applied monotonicity (CVaR is nondecreasing in portfolio variance at fixed mean) to reduce the mean-CVaR-skewness problem to a minimum-variance program, and showed numerically that lower variance also raises skewness · [\[Poster\]](#)

## EMPLOYMENT

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### Grader

Department of Statistics, Columbia University

Feb 2026 – Present

- Courses: STAT GR5264 Stochastic Processes – Applications I (graduate) and STAT GU4001 Introduction to Probability and Statistics
- Graded assignments and exams for proof-based and applied coursework, and coordinated with instructors on grading standards and rubric consistency

### Intern

Business Department, Bank of China

Jul 2022 – Aug 2022

- Used SQL to analyze customer transaction records, screening high-risk transactions and flagging potential money-laundering activity

## SKILLS

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**Python** (NumPy, pandas, SciPy, scikit-learn, statsmodels, GeoPandas, Shapely, OSMnx) ·

**R** (lme4, lmerTest; REML mixed-effects models) · **Stan** (CmdStanR) · **SQL** (DuckDB) · **LaTeX**

## HONORS & AWARDS

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- **First-Class Honours**, Bachelor of Science in Financial Mathematics, University of Liverpool (2025)
- **University Academic Achievement Award**, XJTLU (2024/25)
- **Summer Undergraduate Research Fellowship (SURF)**, XJTLU (2024)

## EXTRACURRICULAR & VOLUNTEER

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- **Archery Club, XJTLU**: Director, Equipment Maintenance Department (2022–2023)
- **Jiangsu Province Archery Championship**: Team Bronze, Compound Bow (2023)
- **Red Cross Society of China**: Certified First Aider (2024)